Project Report Requirements Document

Real-Time Financial Fraud Detection using Machine Learning & Stream Processing

CPS 698

Mentor Name: Dr. Sisheng Liang

Team 3

Members: Vinay Kumar Simma, Pradeep Yarlagadda, Praneeth Reddy Ambati

This document outlines the structure and content that will be included in the final project report for the above-titled project. The report will be formatted following the structure of a scientific research paper, as the project involves machine learning and addresses a research-oriented problem. This format has been advised by our mentor.

Repo

The final report will include the following sections:

I. Introduction

- Background and context of financial fraud detection
- Statement of the research problem (latency in large-scale classification)
- Objectives and scope (deploying real-time streaming ML pipelines)
- Significance of the work

II. Methods

- Dataset description and preprocessing techniques (Paysim data)
- Machine learning models/architectures used (XGBoost)
- Real-time Streaming architecture (Apache Kafka, FastAPI, SQLite, Streamlit)
- Training methodology and configuration

III. Results

- Model performance on simulated live streaming data
- Evaluation metrics (accuracy, precision, recall, log-loss)
- System latency and operational throughput

IV. Application

- Real-world deployment scenario on AWS EC2 via Docker
- Potential scalability of the Kafka broker and stream processing
- Implications for the fintech and banking industry

V. Discussion

- Interpretation of the pipeline robustness
- Challenges faced during the project (concurrency locks, environment configurations)
- Limitations and future work
- Lessons learned

Appendices:

The final report will also include the following appendices as required:

- Appendix A: Approved project proposal
- Appendix B: Updated GitHub/Code repository link with documentation
- Appendix C: Weekly meeting minutes
- Appendix D: Final project self-evaluation

References

The report will include in-line citations as needed. We will use the IEEE format for listing all sources and references at the end of the report.

Prop